A

Mini Project Report

on

Multi-Platform Product Review Sentiment Analyzer

Submitted in partial fulfillment of the requirements for the

degree

Third Year Engineering – Computer Science Engineering (Data Science)

by

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ABSTRACT

The rapid growth of e-commerce platforms has led to an increase in the number of product reviews, which play a crucial role in influencing customer purchasing decisions. However, these reviews are often scattered across multiple platforms and available in large volumes, making manual analysis difficult and time-consuming. This project proposes a Multi-Platform Product Review Sentiment Analyzer that utilizes Natural Language Processing (NLP) techniques to automatically classify reviews into positive, negative, and neutral sentiments. The system collects reviews from various sources, identifies the most positive and negative feedback, and presents the results through an interactive dashboard. By providing a consolidated and easy-to-understand summary of customer opinions, the system helps users make faster and more informed purchasing decisions.

Chapter 1

Introduction

The rapid advancement of digital technologies and the widespread availability of the internet have led to significant growth in e-commerce platforms, fundamentally transforming the way people shop for products and services. Platforms such as Amazon, Flipkart, and other online marketplaces provide customers with the convenience of browsing, comparing, and purchasing products from anywhere at any time. One of the most important features of these platforms is the ability for users to share their experiences through product reviews and ratings. These reviews act as a valuable source of information for potential buyers, helping them evaluate product quality, usability, and overall satisfaction before making a purchase decision.

However, with the increasing popularity of online shopping, the number of product reviews has grown exponentially. These reviews are often distributed across multiple platforms, making it difficult for users to gather and analyze them effectively. Manually reading hundreds or even thousands of reviews is not only time-consuming but also impractical. Additionally, users may encounter biased or conflicting opinions, which further complicates the decision-making process. As a result, there is a need for an automated system that can efficiently process and summarize large volumes of review data to provide meaningful insights.

To address this challenge, sentiment analysis, a subfield of Natural Language Processing (NLP), can be utilized to automatically analyze textual data and determine the sentiment expressed in reviews. By classifying reviews into categories such as positive, negative, and neutral, sentiment analysis enables a quick understanding of overall customer opinions. The proposed system, Multi-Platform Product Review Sentiment Analyzer, aims to collect product reviews from multiple e-commerce platforms, process and analyze them using sentiment analysis techniques, and present the results in a structured and user-friendly manner.

Furthermore, the system focuses on identifying the most positive and most negative reviews, which can provide deeper insights into the strengths and weaknesses of a product. The analyzed data is then visualized through an interactive dashboard, allowing users to easily interpret sentiment distribution, ratings, and key feedback trends. This approach not only saves time but also enhances the decision-making process by providing a clear and concise summary of customer opinions. Overall, the system aims to bridge the gap between large-scale review data

and user understanding, enabling customers to make faster, smarter, and more informed purchasing decisions.

1.1 Purpose

The primary purpose of this project is to simplify the process of analyzing product reviews collected from multiple e-commerce platforms. With the increasing number of online reviews, it becomes difficult for users to manually read and interpret all the feedback before making a purchase decision. This system aims to overcome this challenge by providing an automated solution that can efficiently process large volumes of review data.

Another key purpose is to utilize sentiment analysis techniques from Natural Language Processing (NLP) to classify reviews into positive, negative, and neutral categories. By doing so, the system helps in understanding the overall sentiment of customers toward a product without requiring manual effort. It also identifies the most positive and most negative reviews, offering deeper insights into product strengths and weaknesses.

Furthermore, the project aims to present the analyzed data in a clear and visually appealing manner through an interactive dashboard. This enables users to quickly interpret review trends, sentiment distribution, and ratings. Ultimately, the purpose of this system is to assist users in making faster, smarter, and more informed purchasing decisions by providing a consolidated and easy-to-understand summary of customer opinions.

1.2 Problem Statement

The problem statement for the Multi-Platform Product Review Sentiment Analyzer revolves around the challenges faced by users in analyzing large volumes of product reviews available across multiple e-commerce platforms. While reviews play a crucial role in influencing purchasing decisions, existing methods fail to provide a unified, efficient, and meaningful summary of customer opinions. Users are often required to manually read and interpret numerous reviews, leading to inefficiencies, confusion, and incomplete understanding of a product’s overall perception.

Key challenges that this system seeks to address include:

1. **Lack of Integration:** Product reviews are scattered across different platforms, making it difficult to obtain a consolidated and unified view of customer feedback.

2. **Time Constraints:** Users often do not have sufficient time to read and analyze hundreds of reviews before making a purchase decision.

3. **Information Overload**: The large volume of reviews can overwhelm users, making it difficult to identify the most relevant and useful feedback.

4. **Lack of Clear Sentiment Insight:** Existing systems do not effectively summarize whether overall feedback is positive, negative, or neutral.

5. **Difficulty in Identifying Key Reviews:** Users struggle to find the most helpful positive and negative reviews that highlight product strengths and weaknesses.

6. **Manual Analysis Limitations:** Human analysis of reviews is prone to bias, inconsistency, and inefficiency.

7. **Lack of Visualization:** Most platforms do not provide an interactive and visual representation of review insights for quick understanding.

8.

1.3 Objectives

The primary objective of this project is to develop a system that simplifies the analysis of product reviews collected from multiple e-commerce platforms. By utilizing sentiment analysis techniques, the system aims to process large volumes of review data and present meaningful insights in a structured manner. This enables users to better understand customer opinions and supports them in making informed purchasing decisions.

1. To Collect Product Reviews from Multiple E-commerce Platforms

The primary objective of the system is to gather customer reviews of a particular product from various online shopping platforms such as Amazon, Flipkart, Myntra, and others. Instead of visiting each website separately, users can access all reviews in one place. This creates a unified and comprehensive dataset that includes opinions from different sources, helping users get a broader understanding of the product's quality, performance, and customer satisfaction.

2. To Identify and Extract the Best and Worst Reviews

Another important objective is to analyze collected reviews and identify the most positive and most negative customer feedback. Positive reviews highlight product strengths such as durability, design, performance, and value for money, while negative reviews reveal common issues like defects, poor quality, or delivery problems. By presenting the best and worst reviews separately, the system saves users time and allows them to quickly understand the product's advantages and disadvantages.

3. To Visualize Sentiment Distribution and Review Insights

The system aims to present review analysis results in a clear and interactive manner through dashboards, charts, and graphs. Sentiment analysis is used to classify reviews as positive, negative, or neutral. Pie charts, bar graphs, rating trends, and comparison charts help users easily interpret customer opinions. Visual representation makes large amounts of review data more understandable and user-friendly.

4. To Help Users Make Informed Purchasing Decisions

The final objective is to support customers in making smart buying decisions. By providing summarized insights, sentiment trends, and genuine customer feedback from multiple platforms, the system reduces confusion and saves time. Users can compare opinions, identify product reliability, and purchase with greater confidence. This leads to faster, better, and more informed decision-making.

1.4 Scope

The scope of the Multi-Platform Product Review Sentiment Analyzer is to develop a system that collects, analyzes, and summarizes product reviews from multiple e-commerce platforms, providing users with clear and actionable insights. The system is designed to simplify decision-making by transforming large volumes of textual data into meaningful sentiment-based information.

The specific scope of the project is as follows:

1. **Target Users:**

To serve online shoppers, consumers, and researchers by providing an easy-to-use platform for understanding product reviews and customer opinions.

2. **Feature Integration:**

To provide a unified system that includes review collection, sentiment classification (positive, negative, neutral), identification of key reviews, and visualization through an interactive dashboard.

3. **Technology Implementation:**

To utilize technologies such as Natural Language Processing (NLP), machine learning models (Python), and web technologies for dashboard visualization and user interaction.

4. **Scalability:**

To design the system in a way that it can handle increasing volumes of review data and support integration with multiple e-commerce platforms in the future.

5. **Limitations:**

To acknowledge that sentiment analysis accuracy depends on the quality of review data and language complexity, and that the system may not always perfectly interpret sarcasm, mixed sentiments, or context-specific opinions.

Chapter 2

Literature Review

The literature survey focuses on analyzing existing research and methodologies related to sentiment analysis of product reviews. Various studies have explored the use of Natural Language Processing (NLP) and machine learning techniques to classify and interpret customer feedback from e-commerce platforms. These works highlight different approaches for extracting insights, improving accuracy, and handling large volumes of review data.

The review of these research papers provides a strong foundation for the proposed Multi-Platform Product Review Sentiment Analyzer, helping to understand current advancements, identify limitations, and design a more efficient and comprehensive system.

Several research works have been carried out in the field of sentiment analysis for product reviews, focusing on extracting meaningful insights from large volumes of customer feedback.

Sr. No	**Title**	**Author(s)**	**Year**	**Outcome**	**Methodology**	**Result**
1.	[1]Sentiment Analysis for Product Reviews	Anil B., Aman Kumar Singh, Aditya Kumar Singh, Surekha Thota	2019	The system analyzes a large number of product reviews automatically and identifies their sentiment polarity as positive, negative, or neutral. This helps users quickly understand overall customer opinion and make better purchasing decisions.	Used NLP techniques and VADER sentiment analysis on reviews collected from Amazon, Flipkart, and Snapdeal.	Demonstrated that sentiment analysis can efficiently classify customer opinions into positive, negative, and neutral sentiments.

Sr. No	Title	Author	Year	Outcome	Methodology	Result
2.	[2] Sentiment Analysis based Method for Amazon Product Reviews	Jyoti Budhwar, Sukhdip Singh	2021	Extracts useful insights from product reviews to help users understand customer feedback	Collected Amazon reviews and applied NLP preprocessing, tokenization, and sentiment classification to determine positive and negative opinions.	The system successfully classified reviews into sentiment categories and improved understanding of customer opinions.
3.	[3] Sentiment Analysis of Flipkart Product Reviews using NLP	S. Kiruthika, U. Sneha Dharshini, K. R. Vaishnavi, R. V. Vishwa Priya	2023	The system understands consumer opinions by analyzing both product ratings and written review text. It combines numerical ratings with customer comments to identify satisfaction levels, preferences, and overall sentiment toward the product.	Collected Flipkart product reviews and used Natural Language Processing for sentiment classification.	Results showed that sentiment analysis helps identify customer perception and product success factors.

4.	[4]Sentiment Analysis of Online Shopping Platform Product Reviews	Soumyadeep Paul, Soham Bhattacharyya, Jishnu Chakraborrt, Nabarun Ray	2025	Predicts sentiment polarity of reviews to analyze customer satisfaction levels.	Used machine learning models such as Random Forest and Decision Tree, with preprocessing and review score-based labeling.	Machine learning models effectively categorized reviews and improved sentiment prediction accuracy.

Overall, these studies highlight the importance of sentiment analysis in understanding customer opinions and improving decision-making. They demonstrate the effectiveness of NLP and machine learning techniques in processing large-scale review data. These works form the foundation for the proposed system, which aims to integrate multi-platform data collection, sentiment classification, and visualization for better user insights.

Chapter 3

Proposed System

The proposed system, **Multi-Platform Product Review Sentiment Analyzer**, is designed to provide an automated and efficient solution for analyzing product reviews collected from multiple e-commerce platforms. The system utilizes sentiment analysis techniques from Natural Language Processing (NLP) to classify and interpret customer feedback. By eliminating the need for manual review analysis, it helps users save time and gain a clear understanding of overall product sentiment. The system processes large volumes of data and presents the results in a structured and user-friendly manner, making it easier for users to make informed purchasing decisions.

Key Features of the Proposed System

1. **Multi-Platform Review Collection:**

The system collects product reviews from various e-commerce platforms such as Amazon, Flipkart, and others. This ensures that users receive a comprehensive and unified dataset instead of relying on reviews from a single source, leading to a more accurate and balanced analysis.

2. **Data Preprocessing:**

Before analysis, the collected reviews are cleaned and processed to remove noise such as special characters, punctuation, stop words, and irrelevant information. This step improves the quality of the data and enhances the accuracy of the sentiment analysis model.

3. **Sentiment Analysis:**

The system applies Natural Language Processing (NLP) and machine learning techniques to classify each review into positive, negative, or neutral categories. This allows users to quickly understand the overall opinion about a product without reading every review individually.

4. **Identification of Key Reviews:**

The system identifies and extracts the most positive and most negative reviews from the dataset. These key reviews highlight the major strengths and weaknesses of a product, enabling users to focus on the most important feedback.

5. **Data Visualization:**

The analyzed data is presented through an interactive dashboard that includes charts, graphs, and visual indicators. This helps users easily interpret sentiment distribution, rating trends, and overall insights in a clear and concise manner.

6. **User-Friendly Interface:**

The system is designed with a simple and intuitive interface, allowing users to interact with the dashboard, view results, and access insights without requiring technical knowledge.

7. **Scalability and Future Enhancements:**

The system is built to handle increasing volumes of data and can be extended to support more e-commerce platforms, real-time data collection, and advanced machine learning models. Future improvements may include multilingual sentiment analysis and more accurate prediction capabilities.

By implementing these features, the Multi-Platform Product Review Sentiment Analyzer aims to provide a comprehensive, efficient, and user-friendly solution for analyzing product reviews, enabling users to make smarter and more informed purchasing decisions based on clear and consolidated insights.

3.1 Features & Functionality

The Multi-Platform Product Review Sentiment Analyzer is designed to provide a complete and efficient solution for collecting, analyzing, and presenting product review data. The system integrates multiple features that work together to transform raw review data into meaningful insights, enabling users to easily understand customer opinions and make informed decisions.

1. **Multi-Platform Review Collection:**

The system is capable of collecting product reviews from multiple e-commerce platforms such as Amazon, Flipkart, and others. This feature ensures that users receive a comprehensive dataset instead of relying on a single source. By aggregating reviews from different platforms, the system provides a more accurate and balanced understanding of customer feedback, reducing bias and improving reliability.

2. **Data Preprocessing**

Before performing sentiment analysis, the collected reviews undergo a preprocessing stage. In this step, unnecessary elements such as special characters, HTML tags, punctuation, and stop words are removed. The text is also normalized through techniques like tokenization and lowercasing. This process improves the quality of the data and ensures that the sentiment analysis model works more accurately and efficiently.

3. **Sentiment Classification**

The core functionality of the system lies in its ability to classify reviews based on sentiment. Using Natural Language Processing (NLP) techniques and machine learning algorithms, each review is categorized as positive, negative, or neutral. This automated classification allows users to quickly grasp the general opinion about a product without manually reading every review.

4. **Identification of Key Reviews**

The system identifies and highlights the most positive and most negative reviews from the dataset. These key reviews provide valuable insights into the strengths and weaknesses of a product. By focusing on extreme opinions, users can better understand what aspects of the product are highly appreciated or criticized.

5. **Interactive Dashboard**

The analyzed results are presented through an interactive dashboard that includes visual elements such as bar charts, pie charts, and graphs. These visualizations display sentiment distribution, rating trends, and other insights in a clear and concise manner. The dashboard enhances user experience by making complex data easy to understand at a glance.

6. **Search and Product Selection**

The system includes a search functionality that allows users to select or input a specific product for analysis. Once a product is selected, the system retrieves relevant reviews and displays the sentiment analysis results. This feature makes the system practical and user-friendly for real-time usage.

7. **Consolidated Summary of Reviews**

The system generates a summarized view of all collected reviews by combining sentiment results and key insights. This summary helps users quickly understand overall customer feedback without going through individual reviews, saving time and effort.

8. **Scalability and Future Enhancements**

The system is designed with scalability in mind, allowing it to handle increasing volumes of data as more reviews are collected. It can be extended to include additional features such as real-time data collection, multilingual sentiment analysis, integration with more platforms, and advanced machine learning models for improved accuracy.

Overall, these features and functionalities work together to create a robust and user-friendly system that simplifies the process of analyzing product reviews and enhances decision-making for users.

Chapter 4

Requirements Analysis

Requirement analysis is a crucial phase in the development of the Multi-Platform Product Review Sentiment Analyzer, as it defines the system's functionality, performance expectations, and operational constraints. It helps in understanding what the system should do and how well it should perform. The requirements are broadly categorized into functional and non-functional requirements, along with hardware and software requirements.

Functional Requirements:

Functional requirements describe the specific operations and tasks that the system must perform to achieve its objectives.

1. Review Collection from Multiple Platforms:

The system must be capable of collecting product reviews from various e-commerce platforms such as Amazon, Flipkart, and others. This ensures that users receive a comprehensive and unified dataset for analysis. The system should support data extraction through APIs or web scraping techniques.

2. Data Preprocessing and Cleaning:

The collected data must be preprocessed to remove unwanted elements such as HTML tags, special characters, punctuation, and stop words. Techniques like tokenization, stemming, and normalization should be applied to prepare the data for accurate sentiment analysis.

3. Sentiment Analysis and Classification:

The system should implement NLP and machine learning techniques to classify each review into positive, negative, or neutral sentiment. This classification is essential for summarizing customer opinions effectively.

4. Identification of Key Reviews:

The system must identify and highlight the most positive and most negative reviews. This helps users quickly understand the major strengths and weaknesses of a product without analyzing all reviews manually.

5. Search and Product Selection:

The system should provide a search feature that allows users to enter a product name or select a product for analysis. Based on the input, the system retrieves relevant reviews and displays the results.

6. Data Visualization and Dashboard:

The system must present the analysis results through an interactive dashboard. It should include visual elements such as pie charts, bar graphs, and trend lines to represent sentiment distribution and ratings.

7. Summary Generation:

The system should generate a concise summary of customer opinions based on the analyzed data. This summary should highlight the overall sentiment and key insights.

Non-Functional Requirements:

Non-functional requirements define how the system performs and the quality attributes it must satisfy.

1. Performance:

The system should be capable of processing large volumes of review data efficiently and delivering results within a reasonable time frame.

2. Scalability:

The system should be designed to handle increasing data loads and support the addition of more e-commerce platforms in the future without performance degradation.

3. **Usability:**

The user interface should be simple, intuitive, and easy to navigate so that users with minimal technical knowledge can operate the system effectively.

4. **Accuracy:**

The sentiment analysis model should provide accurate and reliable classification results, minimizing errors in interpretation.

5. **Reliability:**

The system should operate consistently without crashes or failures, ensuring continuous availability to users.

6. **Security:**

Proper security measures should be implemented to protect data and prevent unauthorized access.

Chapter 5

Project Design

5.1 Case Diagram:

Figure 5.1 presents the Use Case Diagram for the ReviewLens**:** Multi-Platform Product Review Sentiment Analyzer, illustrating the primary interactions between stakeholders and the system's core analytical engine. As depicted in **Figure 5.1**, the diagram highlights how the "Admin" manages model performance and system monitoring, while the "User" interacts with functionalities such as searching product reviews and accessing the sentiment insights dashboard.

The architecture shown in **Figure 5.1** also details the integration of specialized backend modules, where review data is autonomously preprocessed and routed to either Sklearn for traditional classification or OpenRouter for advanced, large-scale NLP analysis. By serving as a structural blueprint, **Figure 5.1** ensures that both the functional requirements—such as automated data storage—and the non-functional requirements—such as high-speed dashboard generation via FastAPI—are seamlessly addressed within the system's design.

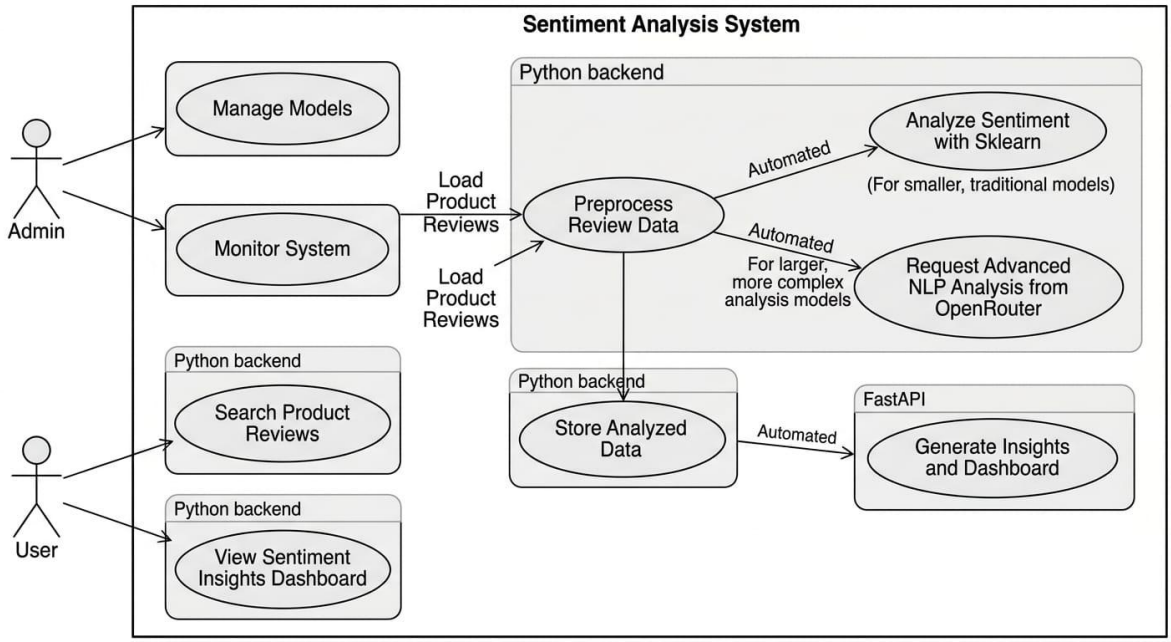

Figure 5.1 : Case Diagram for Quantamental

5.2 DFD (Data Flow Diagram):

Figure 5.2.1 presents the **Data Flow Diagram (DFD) Level 0**, also known as the Context Diagram, for the **ReviewLens** sentiment analysis system. As illustrated in **Figure 5.2.1**, the diagram centralizes the **0.0 Sentiment Analysis** process as the core engine that manages data exchanges between external entities and the internal database.

The diagram in **Figure 5.2.1** highlights how the "Admin" provides system configurations and model parameters, while the "User" interacts with the system through search queries and dashboard requests. Furthermore, **Figure 5.2.1** details the critical flow of **Raw Product Reviews** from the data store to the processing engine, followed by the exchange of NLP requests and responses with the **OpenRouter AI** external entity. By serving as a high-level data blueprint, **Figure 5.2.1** ensures that the boundaries and major data interfaces of the multi-platform analyzer are clearly defined within the technical documentation.

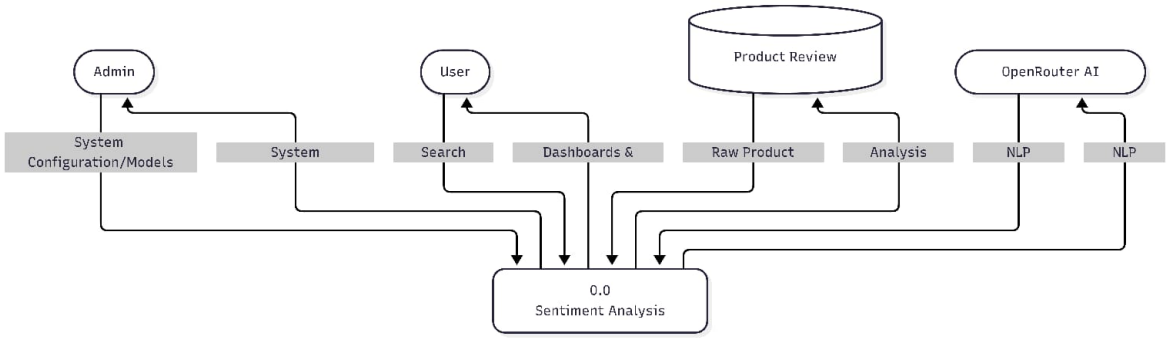

Figure 5.2.1: Level [0] Data Flow Diagram of *ReviewLens* System

Figure 5.2.2 presents the **Data Flow Diagram (DFD) Level 1**, providing a detailed breakdown of the internal processes that drive the **ReviewLens** sentiment analysis system. As illustrated in **Figure 5.2.2**, the high-level "Sentiment Analysis" process is decomposed into five distinct functional modules, each utilizing specific technologies like **Pandas, Scikit-learn, and FastAPI**.

The diagram in **Figure 5.2.2** highlights the modularity of the **Data Pipeline & Core**, where raw reviews are transformed through a preprocessing stage (2.0) before being routed to either local sentiment classification (3.0) or complex remote NLP analysis via OpenRouter (4.0). Furthermore, Figure 5.2.2 demonstrates the lifecycle of data from the initial **Admin** configuration to the final generation of user insights (5.0), ensuring a clear path from raw

data ingestion to the delivery of visual dashboards. By defining these sub-processes, Figure 5.2.2 acts as a technical guide for understanding the system's internal logic and technology stack integration.

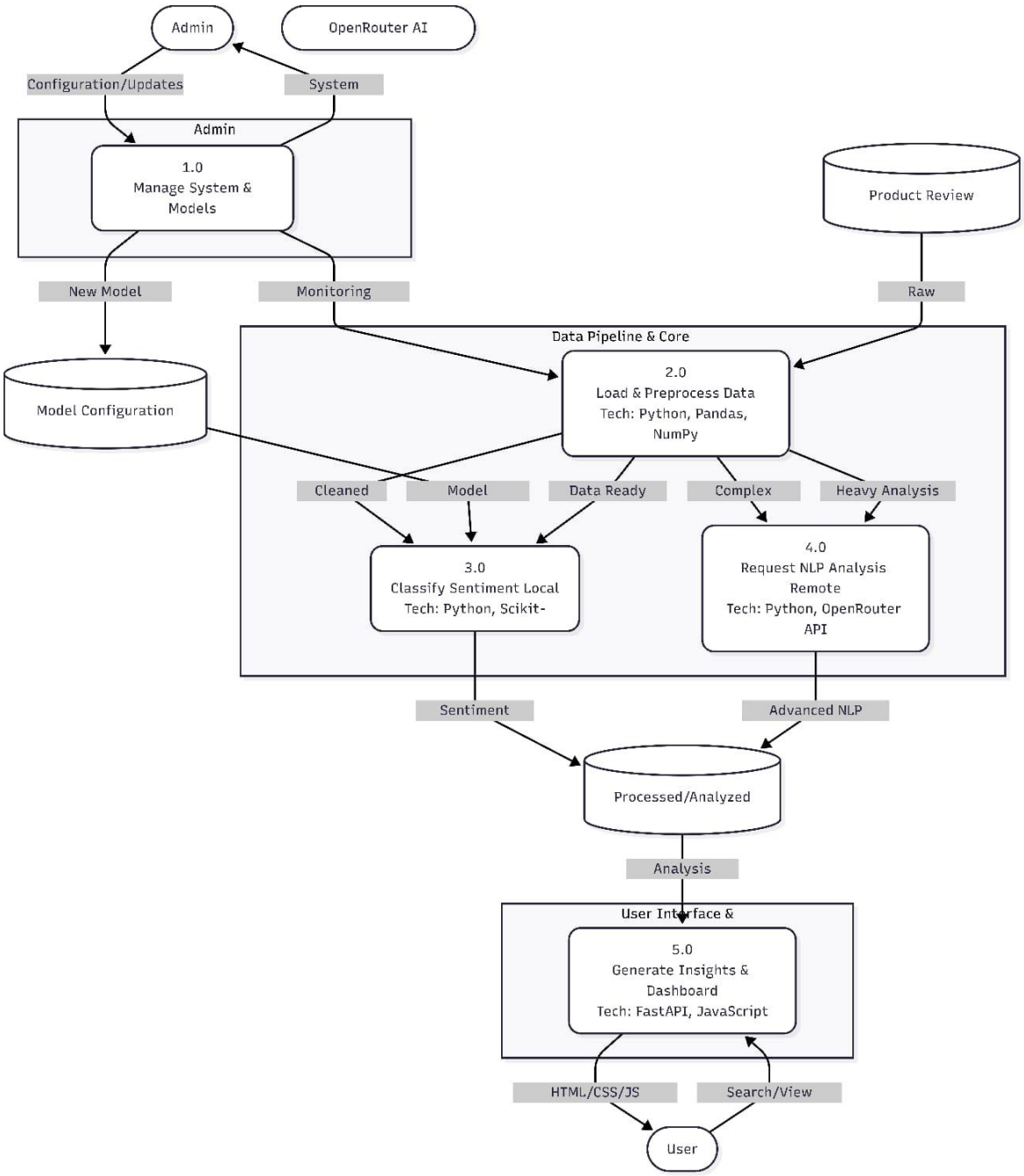

Figure 5.2.2: Level [1] Data Flow Diagram of ReviewLens

Figure 5.2.3 provides a high-resolution functional decomposition in the form of a **Data Flow Diagram (DFD) Level 2**, specifically detailing the sub-processes within the **Load & Preprocess Data** module (Process 2.0). As illustrated in **Figure 5.2.3**, the system transforms raw, unstructured data into machine-readable formats through a rigorous four-stage pipeline. The diagram in **Figure 5.2.3** tracks the data journey beginning with **Process 2.1**, where raw reviews are fetched based on Admin filtering criteria. This is followed by **Process 2.2** (Text Cleaning using Regex/NLTK) and **Process 2.3** (Tokenization), which refine the text into its most meaningful components. Finally, **Figure 5.2.3** shows **Process 2.4** (Vectorization), which converts tokens into a feature matrix or text payload, determining whether the data is routed to local classification (3.0) or remote NLP analysis (4.0). By isolating these granular steps, **Figure 5.2.3** ensures that the data engineering phase of the Multi-Platform Product Review Sentiment Analyzer is both transparent and highly efficient.

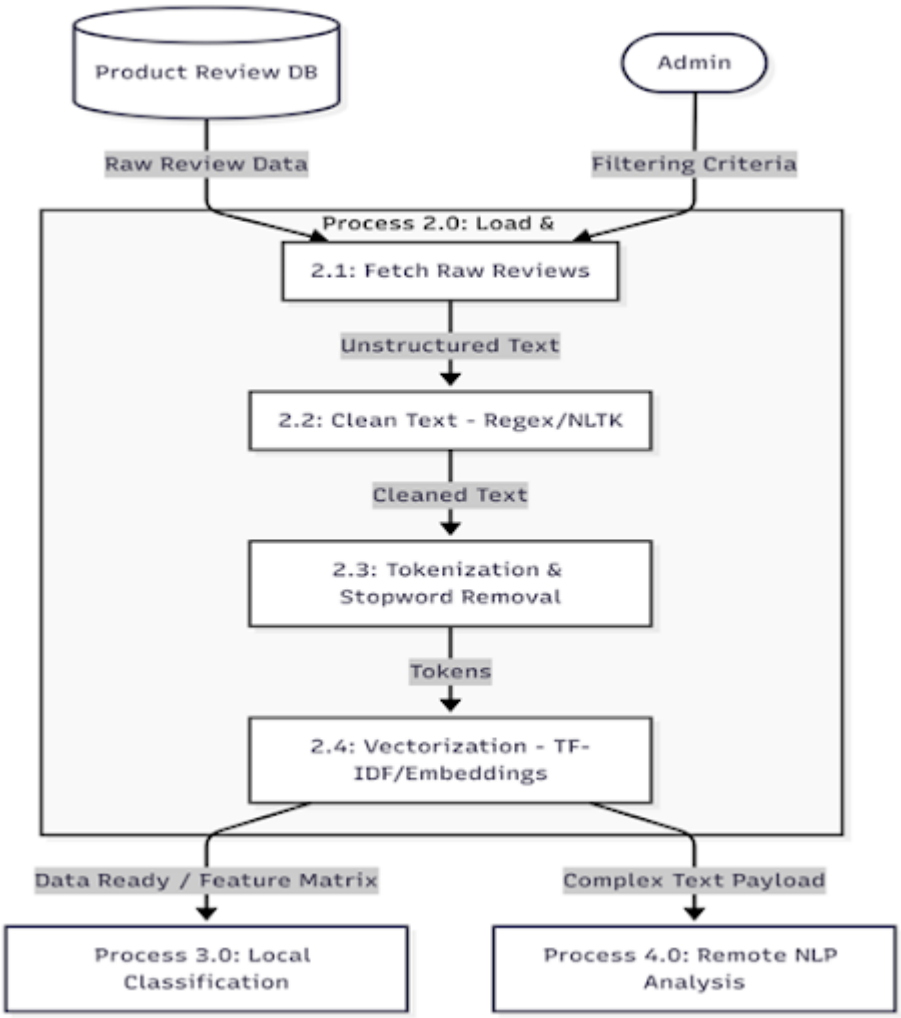

Figure 5.2.3: Level [2] Data Flow Diagram of ReviewLensSyste

5.3 System Architecture:

Section 5.3 details the **System Architecture** of the **ReviewLens** platform, providing a high-level overview of the structural design and the interaction between its multi-tier components. As illustrated in the architecture diagram, the system is built on a modular framework that separates the **Presentation Tier** (Web UI) from the **Application Tier** (Python Backend), ensuring a decoupled and scalable environment.

The architecture highlights a hybrid intelligence approach, where a centralized **Data Orchestration & Routing** engine directs preprocessed reviews to either local **Scikit-learn** models for high-speed classification or to the **OpenRouter API** for nuanced, context-aware NLP analysis. By integrating external data sources from multiple e-commerce platforms and maintaining a structured **Data Tier** for both raw and analyzed information, the architecture ensures that the system can handle large-scale sentiment intelligence tasks with maximum accuracy and minimal latency.

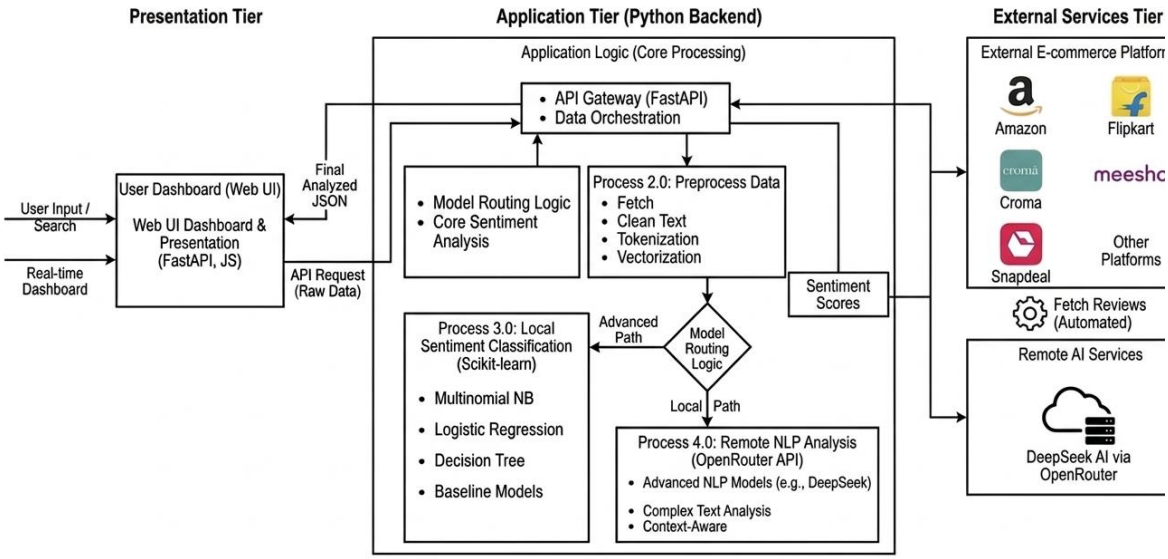

Figure 5.3: System Architecture Quantamental System

5.4 Implementation:

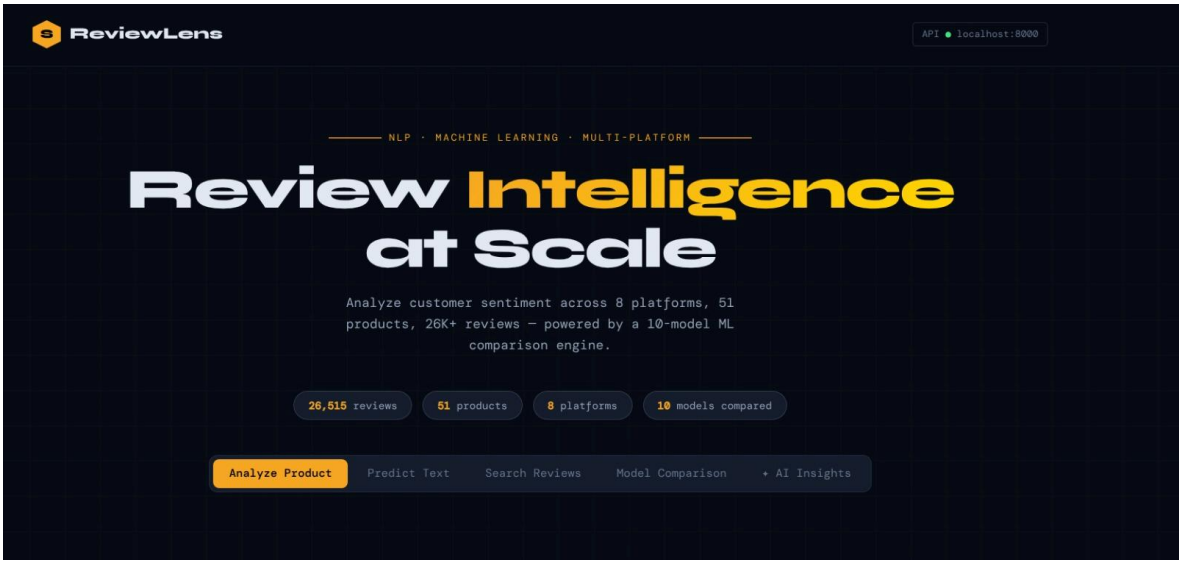

Figure 5.4.1:Landing page

Figure 5.4.1 presents the **ReviewLens** dashboard, which serves as the primary interface for large-scale sentiment analysis and data monitoring. This page, as illustrated in **Figure 5.4.1**, leverages a sophisticated **10-model Machine Learning** comparison engine to process over 26,000 customer reviews across eight distinct platforms. By centering the user experience on "Review Intelligence," **Figure 5.4.1** highlights how the system translates raw feedback into structured data through integrated **NLP** modules and real-time API connectivity.

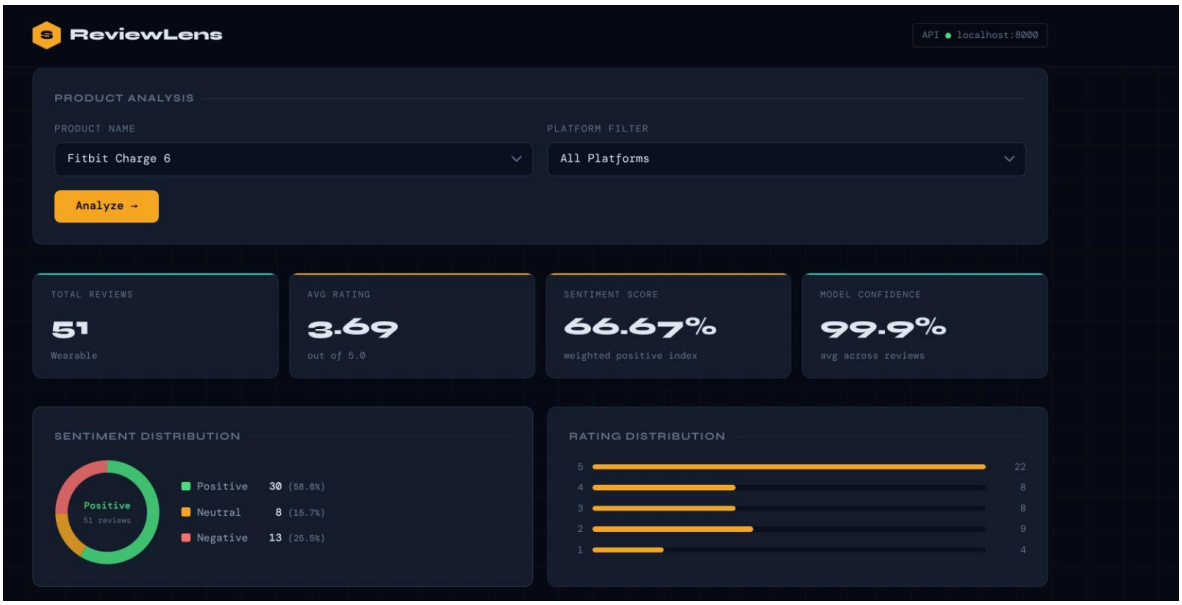

Figure 5.4.2:Product Analysis

Figure 5.4.2 illustrates the detailed "Product Analysis" view within the ReviewLens platform, specifically showcasing the granular data breakdown for a selected item, the Fitbit Charge 6. As depicted in Figure 5.4.2, this interface provides essential performance metrics including a Sentiment Score of 66.67% and a remarkably high Model Confidence of 99.9%. The visual data representation in Figure 5.4.2 further categorizes customer feedback through a sentiment distribution donut chart and a star-rating frequency bar chart, allowing for a comprehensive evaluation of consumer reception across all integrated platforms.

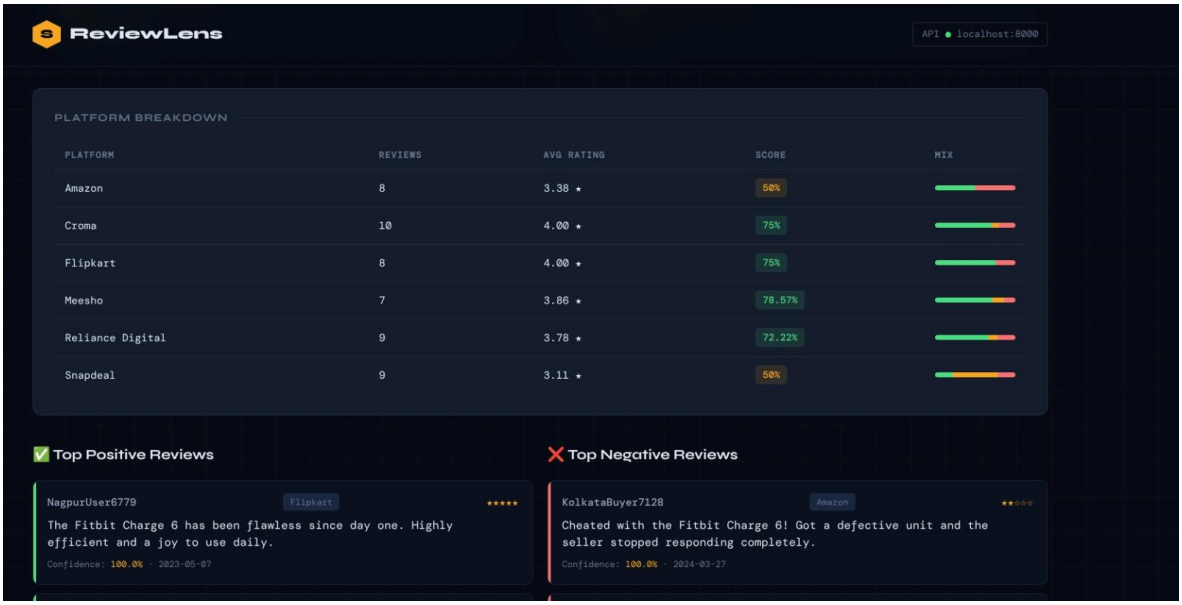

Figure 5.4.3:Platform Breakdown

Figure 5.4.3 provides a comprehensive **Platform Breakdown** and sentiment deep-dive, offering a comparative analysis of how a product performs across various e-commerce channels. As shown in **Figure 5.4.3**, the system aggregates data from multiple platforms—such as Amazon, Flipkart, and Croma—calculating a specific **Average Rating** and **Score** for each. Furthermore, **Figure 5.4.3** highlights the qualitative aspect of the analysis by surfacing "Top Positive Reviews" and "Top Negative Reviews," complete with individual model confidence levels and source attribution. This multi-platform granularity allows for the identification of specific logistics or regional feedback trends that might be obscured in a global average.

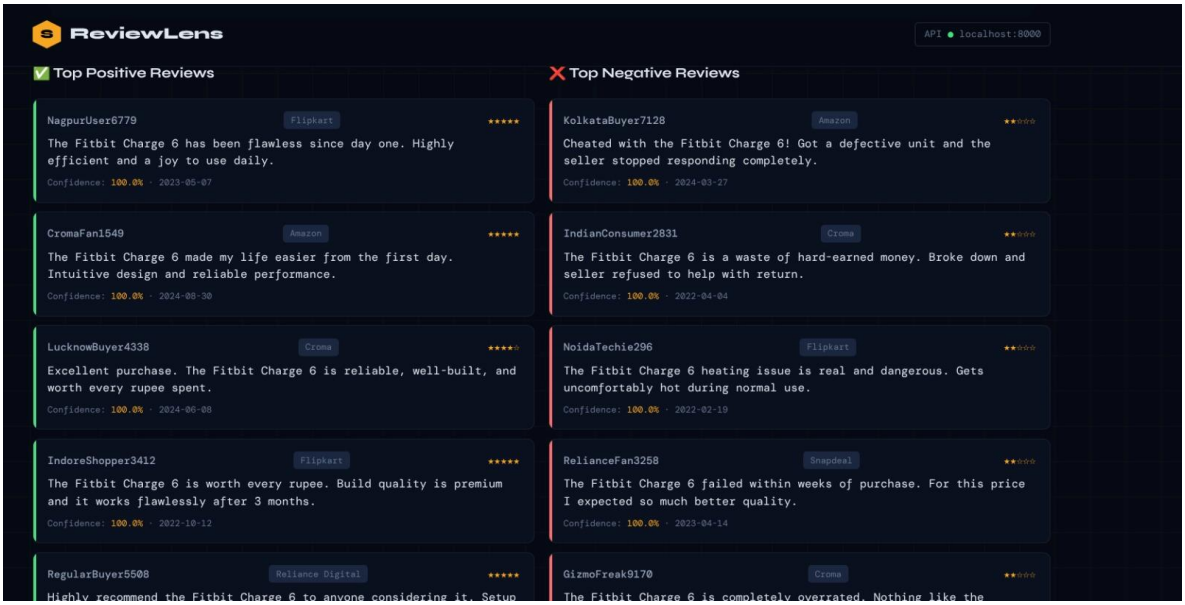

Figure 5.4.4: Positive and Negative Reviews module

Figure 5.4.4 presents a granular view of the qualitative data through the Top Positive and Negative Reviews module. As illustrated in Figure 5.4.4, the interface lists verbatim customer feedback side-by-side to allow for a direct thematic comparison between user satisfaction and common pain points. Each review card in Figure 5.4.4 includes critical metadata, such as the source platform (e.g., Flipkart, Amazon, Croma), the star rating, a date stamp, and a 100.0% Model Confidence score, indicating the AI's high certainty in its sentiment classification. This specific view in Figure 5.4.4 is vital for identifying recurring issues, such as the "heating issue" mentioned in the negative column, which provides actionable intelligence beyond simple numerical averages.

Chapter 6

Technical Specification

The Multi-Platform Product Review Sentiment Analyzer is developed using a modern and efficient technology stack to ensure high performance, scalability, and accurate sentiment analysis. The system integrates backend processing, AI services, and frontend technologies to deliver a seamless user experience.

Technology Stack:

- **Python:**
Python is used as the core programming language due to its simplicity and strong support for data analysis and machine learning. It is primarily used for implementing sentiment analysis and backend logic.

- **FastAPI:**
FastAPI is used to build high-performance APIs for handling requests, processing data, and connecting the frontend with the backend. It ensures fast execution and easy integration with AI models.

- **NumPy:**
NumPy is used for numerical computations and efficient handling of large datasets, which is essential for processing review data.

- **Pandas:**
Pandas is used for data manipulation and analysis, including cleaning, organizing, and structuring review datasets.

- **Scikit-learn (Sklearn):**

Scikit-learn is used to implement machine learning models for sentiment classification, including training and evaluating models for accurate predictions.

1. **AI Service:**

- **OpenRouter:**
OpenRouter is used as the AI service to access advanced language models for performing sentiment analysis and generating insights. It helps in improving accuracy and handling complex language patterns in reviews.

2. **Frontend Technologies**

- **HTML:**
HTML is used to structure the web interface and display content such as review summaries and dashboard elements.

- **CSS:**

CSS is used to design and style the user interface, making it visually appealing and user-friendly.

- **JavaScript:**

JavaScript is used to add interactivity, handle user inputs, and dynamically update the dashboard with analysis results.

3. System Integration

The frontend communicates with the backend through FastAPI endpoints. The backend processes the data using Python libraries and machine learning models, while OpenRouter enhances AI-based analysis. The final results are sent back to the frontend and displayed through an interactive dashboard.

This technology stack ensures that the system is efficient, scalable, and capable of delivering accurate sentiment analysis with a smooth user experience.

Chapter 7

Project Scheduling

In project management, a schedule is a listing of a project's milestones, activities, and deliverables. A schedule is commonly used in project planning and project portfolio management. The project schedule (Table 7.1) links the tasks to be performed with the group members responsible for completing them and the time duration allocated.

Sr.no	Group Members	Duration	Task Performed
1.	Gaurav Shetty Shlok Sawant Chitresh Poojary Manish Sharma	2nd Week of January	Group formation and topic finalization. Identifying the scope and objectives of the project "Multiplatform product review sentiment analyzer." Discussing the project idea and preparing a basic paper prototype.
		1st Week of January	Researching existing investment analysis platforms and identifying functional requirements for Multiplatform product review sentiment analyzer.
2.	Chitresh Poojary Shlok Sawant Manish Sharma	2nd Week of February	Developing the backend structure using Python and FastAPI for handling APIs, processing product reviews, and managing server-side logic.
3.	Manish Sharma Chitresh Poojary	3rd Week of February	Implementing AI/ML models using **Scikit-learn**, **NumPy**, and **Pandas** for sentiment analysis and data preprocessing of product reviews.
4.	Gaurav Shetty Shlok Sawant	4th Week of February	Designing the frontend using HTML, CSS, JavaScript, and integrating dashboards to display sentiment insights, review analytics, and interactive visualizations.

5.	Gaurav Shetty Manish Sharma	1st Week of March	Integrating OpenRouter for advanced natural language processing and enhancing sentiment analysis with real-time AI responses.
6.	Gaurav Shetty Chitresh Poojary	2nd Week of March	Integrating all modules and testing the system for functionality, performance, and data consistency.

Table 7.1:Project Task Distribution

A Gantt chart is a type of bar chart that illustrates a project schedule. This chart lists the tasks to be performed on the vertical axis, and time intervals on the horizontal axis. Gantt chart (Fig 7.1) illustrates the start and finish dates of the terminal elements and summary elements of a project

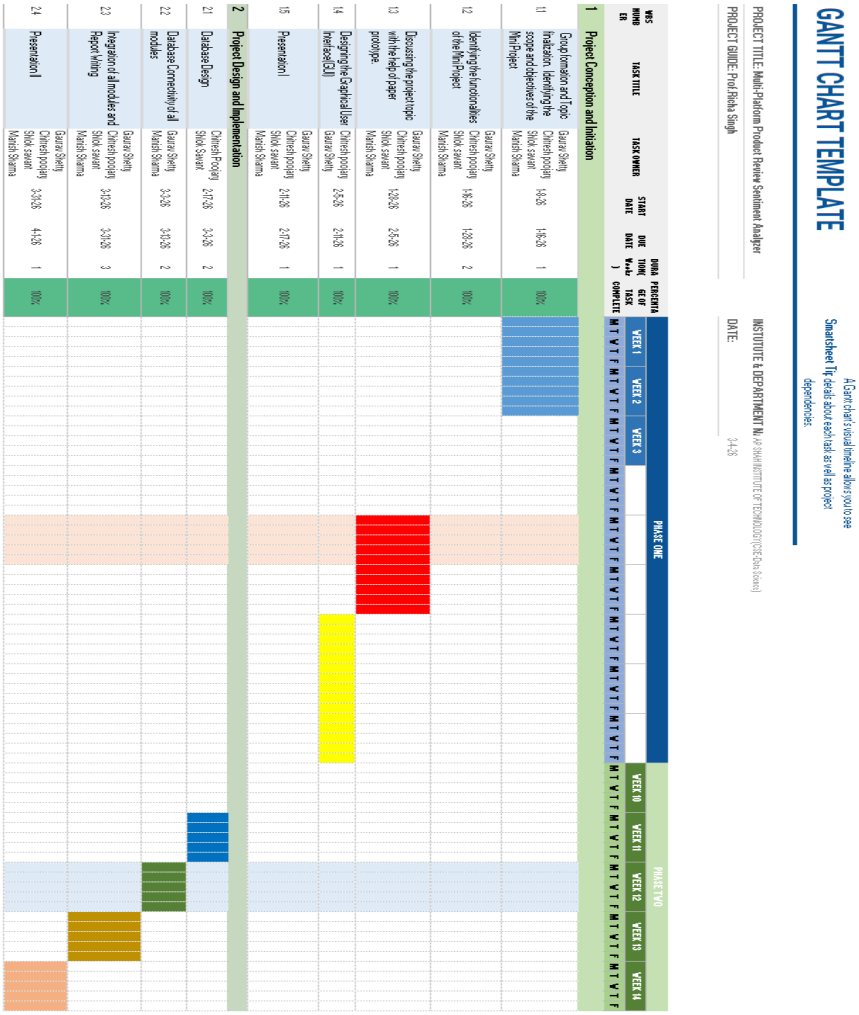

Figure 7.1:Gant Chart

Chapter 8

Results

	MODEL	MACRO F1	ACCURACY	TRAIN RELATIVE
🥇	Multinomial NB BEST	0.9570	0.9605	0.04s
🥈	Logistic Regression	0.9564	0.9625	0.08s
🥉	Complement NB	0.9529	0.9605	0.04s
#4	Ridge Classifier	0.9520	0.9585	0.18s
#5	Linear SVC	0.9518	0.9585	0.19s
#6	SGD Classifier	0.9474	0.9545	0.07s
#7	Decision Tree	0.9456	0.9506	0.11s
#8	Random Forest	0.9411	0.9466	0.23s
#9	KNN	0.9410	0.9486	0.04s
#10	Gradient Boosting	0.9329	0.9407	8.14s

Figure 8.1:Model analysis

Figure 8.1 presents the performance leaderboard of the ten machine learning models utilized by the ReviewLens platform for sentiment classification. As shown in Figure 8.1, each model is evaluated based on its Macro F1 Score, Accuracy, and Relative Training Time (measured in seconds). According to the data in Figure 8.1, the Multinomial Naive Bayes (NB) model is identified as the "Best" performer with a Macro F1 score of 0.9570 and a highly efficient

training time of 0.04s. The comparative table in Figure 8.1 highlights the trade-offs between predictive precision and computational cost, ranging from high-speed classifiers like KNN to more resource-intensive models like Gradient Boosting, which recorded a training time of 8.14s.

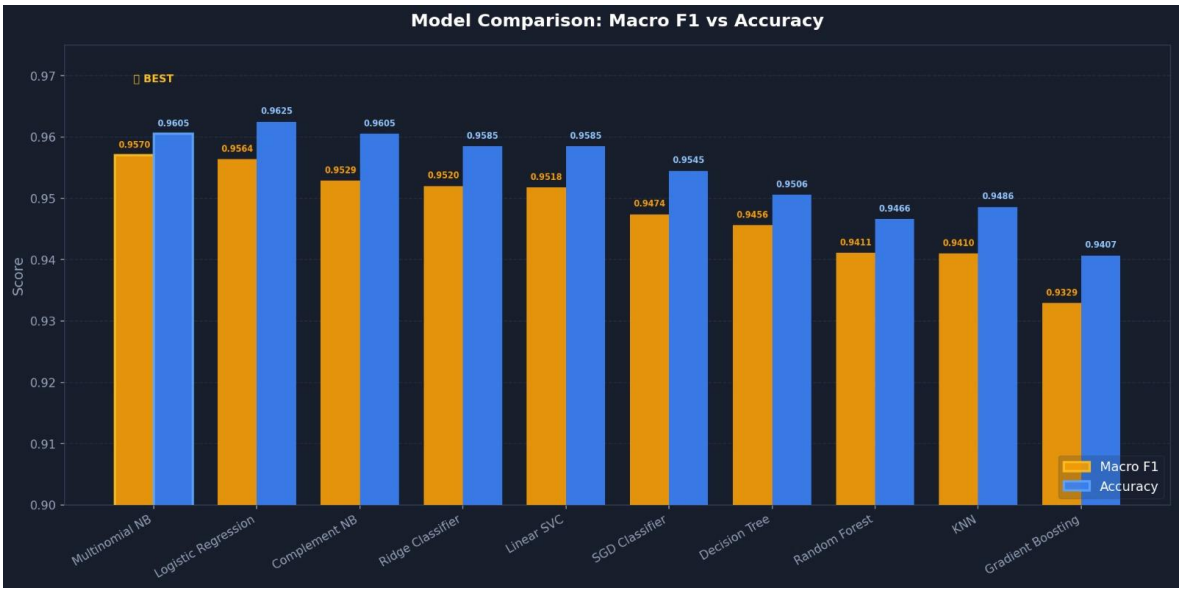

Figure 8.2: visual comparative analysis

Figure 8.2 provides a visual comparative analysis through a grouped bar chart, contrasting the Macro F1 Score against the Accuracy for each of the ten evaluated models. As depicted in Figure 8.2, this visualization allows for a clear assessment of model performance consistency across different metrics. While Accuracy remains relatively high for all models, Figure 8.2 highlights that the Multinomial NB model maintains a superior balance between both metrics, reinforcing its classification as the optimal choice for this specific sentiment analysis task. The downward trend observed from left to right in Figure 8.2 further emphasizes the performance gap between top-tier classifiers and more complex, yet less effective, models like Gradient Boosting in this particular application.

Chapter 9
Conclusion

In conclusion, the Multi-Platform Product Review Sentiment Analyzer successfully addresses the challenges associated with analyzing large volumes of product reviews available across multiple e-commerce platforms. With the increasing dependence of customers on online reviews for decision-making, it becomes essential to have a system that can efficiently process and interpret this vast amount of data. The proposed system leverages Natural Language Processing (NLP) techniques to automate the analysis of textual reviews, thereby eliminating the need for time-consuming manual evaluation.

The system effectively collects reviews from different sources, preprocesses the data to improve quality, and classifies each review into positive, negative, and neutral sentiments. This structured classification provides a clear understanding of overall customer opinions. Additionally, by identifying the most positive and most negative reviews, the system highlights key product strengths and weaknesses, enabling users to focus on the most important feedback. The integration of an interactive dashboard further enhances usability by presenting insights through visual representations such as charts and graphs.

Moreover, the system is designed with scalability and flexibility in mind, allowing it to handle increasing volumes of data and support future enhancements such as real-time analysis and multilingual capabilities. Although the accuracy of sentiment analysis may depend on data quality and language complexity, the system still provides a reliable and efficient approach for summarizing customer feedback.

Overall, the Multi-Platform Product Review Sentiment Analyzer simplifies the process of understanding product reviews, reduces information overload, and enhances decision-making. It serves as a valuable tool for users by providing clear, concise, and meaningful insights, ultimately helping them make faster, smarter, and more informed purchasing decisions.

Chapter 10

Future Scope

The Multi-Platform Product Review Sentiment Analyzer has significant potential for further enhancement and expansion. As technology evolves, the system can be improved to provide more advanced features and better user experience.

1. Real-Time Data Collection

The system can be enhanced to fetch reviews in real-time using APIs or web scraping, ensuring that users always get the latest and most up-to-date information about products.

2. Multilingual Sentiment Analysis

Future versions can support multiple languages, allowing analysis of reviews written in regional and international languages, thereby increasing the system's usability and reach.

3. Advanced Machine Learning Models

The system can integrate advanced deep learning models such as LSTM, BERT, or transformer-based models to improve sentiment analysis accuracy and handle complex language patterns like sarcasm.

4. Fake Review Detection

An additional feature can be implemented to identify and filter fake or spam reviews, improving the reliability and trustworthiness of the analysis.

5. Recommendation System

The system can be extended to suggest similar or better products based on user preferences and sentiment analysis results.

6. Mobile Application Development

A mobile app version of the system can be developed to make it more accessible and convenient for users to analyze reviews on the go.

7. Integration with More Platforms

The system can be expanded to include more e-commerce platforms and even social media sources to provide a broader and more comprehensive analysis.

8. Personalized Insights

Future improvements can include personalized recommendations and insights based on user behavior, preferences, and past interactions.

Overall, the future scope of the project lies in making the system more intelligent, accurate, and user-centric, thereby enhancing its usefulness in real-world applications.

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